

FAGERNES, Norway

WMO: 013670

Lat: 60.9786N

Lon: 9.2239E

Elev: 1199

StdP: 14.07

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.1	-4.9	-14.6	2.6	-9.0	-10.1	3.3	-4.7	14.1	28.2	12.0	19.7	6.6	320	0.616

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.7	77.7	58.7	74.0	57.4	70.5	55.7	61.9	71.7	60.2	69.1	58.4	66.7	5.1	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
58.5	76.4	64.9	56.6	71.3	63.0	54.8	66.7	61.3	28.2	71.9	27.0	69.0	25.9	66.9	68.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
11.9	10.0	9.1	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-12.7	82.0	6.5	3.6	-17.4	84.5	-21.1	86.6	-24.8	88.6	-29.5	91.2		
			-13.4	64.1	6.4	2.4	-18.0	65.8	-21.7	67.2	-25.2	68.6	-29.9	70.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.5	17.0	20.0	28.3	38.0	46.9	55.0	60.3	57.7	49.9	38.9	29.1	19.5		
	DBStd	16.97	11.57	11.51	9.29	5.87	6.35	5.19	5.28	4.78	5.33	6.34	8.21	11.89		
	HDD50	5032	1023	841	672	359	138	13	0	3	65	347	626	945		
	HDD65	9701	1488	1261	1137	809	562	301	166	229	453	809	1076	1410		
	CDD50	834	0	0	0	1	41	164	319	243	63	3	0	0		
	CDD65	26	0	0	0	0	0	3	19	4	0	0	0	0		
	CDH74	311	0	0	0	0	11	58	192	50	0	0	0	0		
	CDH80	35	0	0	0	0	0	6	25	4	0	0	0	0		
	WSAvg	4.0	3.7	3.8	4.2	4.1	4.3	4.2	4.0	3.9	3.8	3.7	3.9	4.5		
	PrecAvg	23.9	1.7	1.1	1.0	1.2	1.9	2.9	3.3	3.4	2.1	2.2	1.9	1.3		
	PrecMax	32.6	3.3	2.2	2.3	3.0	5.8	6.0	7.3	6.4	3.7	5.6	5.6	2.3		
(15) Precipitation	PrecMin	16.2	0.4	0.2	0.2	0.1	0.0	0.3	1.0	0.5	0.8	0.3	0.4	0.2		
	PrecStd	3.5	0.6	0.6	0.6	0.7	1.2	1.3	1.6	1.5	0.7	1.2	1.1	0.6		
	DB	44.6	47.0	52.3	61.8	75.3	79.7	83.2	78.9	70.0	58.9	50.2	45.8			
	MCWB	38.2	38.9	42.6	47.6	55.8	57.9	61.7	58.6	55.8	50.8	43.9	39.5			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	39.5	42.4	47.9	56.6	68.6	74.5	79.2	74.6	64.6	53.6	45.9	41.1			
	2% MCWB	34.5	36.0	39.5	44.2	51.4	55.9	60.4	58.1	53.1	47.3	41.2	37.0			
	5% DB	35.4	38.4	44.0	52.6	63.9	70.3	75.5	70.8	61.5	50.7	41.8	37.8			
	5% MCWB	32.3	33.7	36.5	42.1	49.4	53.5	58.4	57.2	52.1	45.9	38.8	34.5			
	10% DB	32.7	35.1	40.6	48.5	59.1	66.2	71.8	67.2	58.6	48.4	38.8	34.8			
	10% MCWB	30.8	31.7	34.2	39.7	47.3	52.1	57.0	55.9	51.0	44.1	36.4	32.5			
	0.4% WB	38.8	39.5	43.2	49.2	57.6	60.1	65.4	63.3	58.1	52.3	45.4	40.7			
	0.4% MCDB	44.0	45.8	51.4	59.0	69.9	70.7	77.0	71.9	65.0	57.0	49.0	44.7			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	35.1	36.4	40.0	45.3	54.2	58.0	62.9	61.2	55.8	49.0	41.6	37.4			
	2% MCDB	38.6	41.4	47.1	54.2	63.8	69.6	73.7	69.3	61.3	52.0	44.7	40.4			
	5% WB	32.7	34.2	37.0	43.0	51.5	56.2	61.1	59.5	54.2	47.1	39.1	34.8			
	5% MCDB	34.9	38.0	42.9	50.3	60.4	66.4	71.2	66.9	59.3	50.1	41.5	37.1			
	10% WB	30.8	32.1	34.8	41.0	48.9	54.3	59.4	57.8	52.3	44.6	36.6	32.6			
	10% MCDB	32.8	34.7	39.5	47.5	57.7	62.9	68.3	64.3	57.0	47.8	38.6	34.6			
	MDBR	10.7	13.1	16.6	16.6	17.9	17.1	16.7	15.7	14.1	10.1	8.1	9.7			
	5% DB	14.0	15.6	18.4	23.8	27.7	25.9	23.8	22.0	18.7	13.4	12.2	13.4			
(36) Mean Daily Temperature Range	5% DB	11.1	11.1	11.5	12.8	12.8	10.9	9.0	8.5	9.9	8.6	9.5	10.5			
	5% WB	13.0	14.7	17.6	21.3	22.1	19.6	19.4	17.5	15.3	11.2	11.1	12.6			
	MCDBR	10.7	10.9	11.4	12.2	11.3	9.5	8.5	8.6	9.2	8.3	9.1	10.2			
	MCWBR															
(40) Clear-Sky Solar Irradiance	taub	0.219	0.265	0.300	0.339	0.346	0.342	0.357	0.348	0.314	0.297	0.236	0.187			
	taud	2.137	2.248	2.229	2.241	2.394	2.462	2.434	2.448	2.530	2.466	2.205	2.016			
	Ebn at Noon	173	232	260	270	279	282	273	266	259	222	163	129			
	Edh at Noon	12	20	29	35	33	31	32	29	22	17	11	8			
(44) All-Sky Solar Radiation	RadAvg	97	310	787	1341	1557	1623	1476	1154	751	349	121	57			
	RadStd	17	62	76	162	169	158	177	94	88	56	22	5			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	+1.30	N/A	N/A	N/A	N/A
Regional (56 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page